



FutureXcel — Machine Learning

Title:

Customer Churn Prediction Using Machine Learning

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Prepared by:

Sharif Ullah

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1. ABSTRACT

Customer churn is one of the most critical challenges faced by telecommunication companies in the modern competitive market. Retaining existing customers is significantly more cost-effective than acquiring new ones. Therefore, predicting customer churn in advance can help organizations take preventive actions and improve customer retention strategies.

In this project, a machine learning-based customer churn prediction system is developed using the Telco Customer Churn dataset. The dataset contains customer demographic information, service usage details, and billing-related features. Multiple classification algorithms, including Logistic Regression, Random Forest, and Gradient Boosting, were implemented and evaluated. Among these models, Gradient Boosting achieved the highest accuracy of approximately 79.6% and demonstrated balanced performance in predicting churned customers.

Furthermore, a Streamlit-based web application was developed to deploy the trained model, allowing users to input customer information and obtain real-time churn predictions along with probability and risk levels. The proposed system can assist telecommunication companies in identifying high-risk customers and taking proactive retention measures.

2. INTRODUCTION

In today's digital era, the telecommunication industry has become highly competitive, offering similar services at comparable prices. As a result, customers can easily switch from one service provider to another. This phenomenon is known as customer churn. High churn rates directly affect company revenue and long-term business sustainability.

Customer churn prediction aims to identify customers who are most likely to leave a service in the near future. Traditional approaches often fail to analyze large-scale customer data effectively. Machine learning techniques provide data-driven solutions that can learn hidden patterns from historical data and generate accurate churn predictions. This project focuses on applying machine learning models to predict customer churn efficiently and deploy the solution through a user-friendly web interface.

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3. PROBLEM STATEMENT

Telecommunication companies experience significant financial losses due to customer churn. Existing manual or rule-based systems are unable to accurately identify customers at risk of leaving. Due to the increasing volume and complexity of customer data, there is a need for an automated and intelligent system that can analyze customer behavior and predict churn with reasonable accuracy.

The problem addressed in this project is the development of a reliable machine learning-based system that can predict customer churn and support decision-making for customer retention strategies.

4. OBJECTIVES OF THE PROJECT

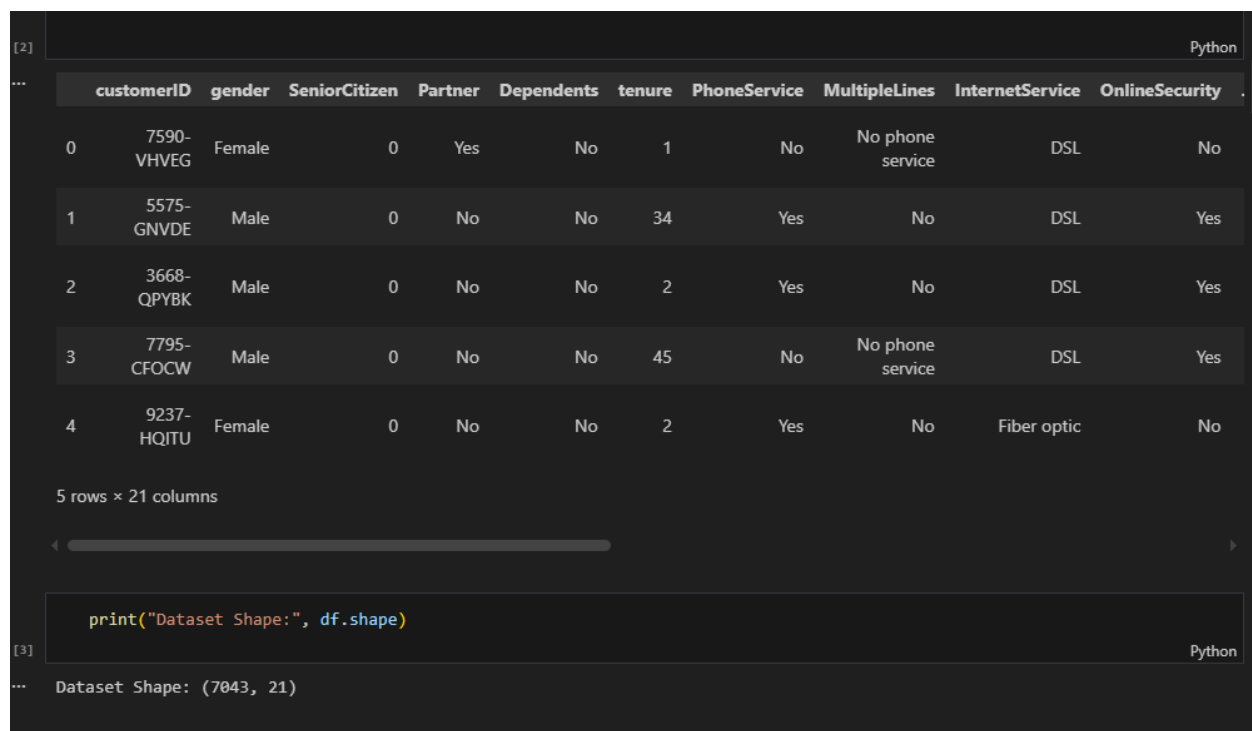
The primary objectives of this project are:

- To understand customer behavior using historical telecommunication data
- To clean and preprocess real-world customer data
- To apply and compare different machine learning classification algorithms
- To evaluate models using appropriate performance metrics
- To select the best-performing churn prediction model
- To develop an interactive web-based churn prediction application

5. DATASET DESCRIPTION

The Telco Customer Churn dataset is a publicly available dataset widely used for churn prediction tasks. The dataset consists of 7,043 customer records and 21 features. These features include customer demographics (gender, senior citizen status), service-related information (internet service, contract type), and billing details (monthly charges, total charges).

The target variable is **Churn**, which indicates whether a customer has left the service (Yes) or not (No). Before modeling, the dataset required preprocessing, including handling missing values and encoding categorical features.



The screenshot shows a Jupyter Notebook interface. The top cell, labeled [2], displays a preview of the dataset with 5 rows and 11 columns. The columns are: customerID, gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetService, and OnlineSecurity. The bottom cell, labeled [3], contains a Python code snippet that prints the dataset shape, which is (7043, 21).

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No

5 rows × 21 columns

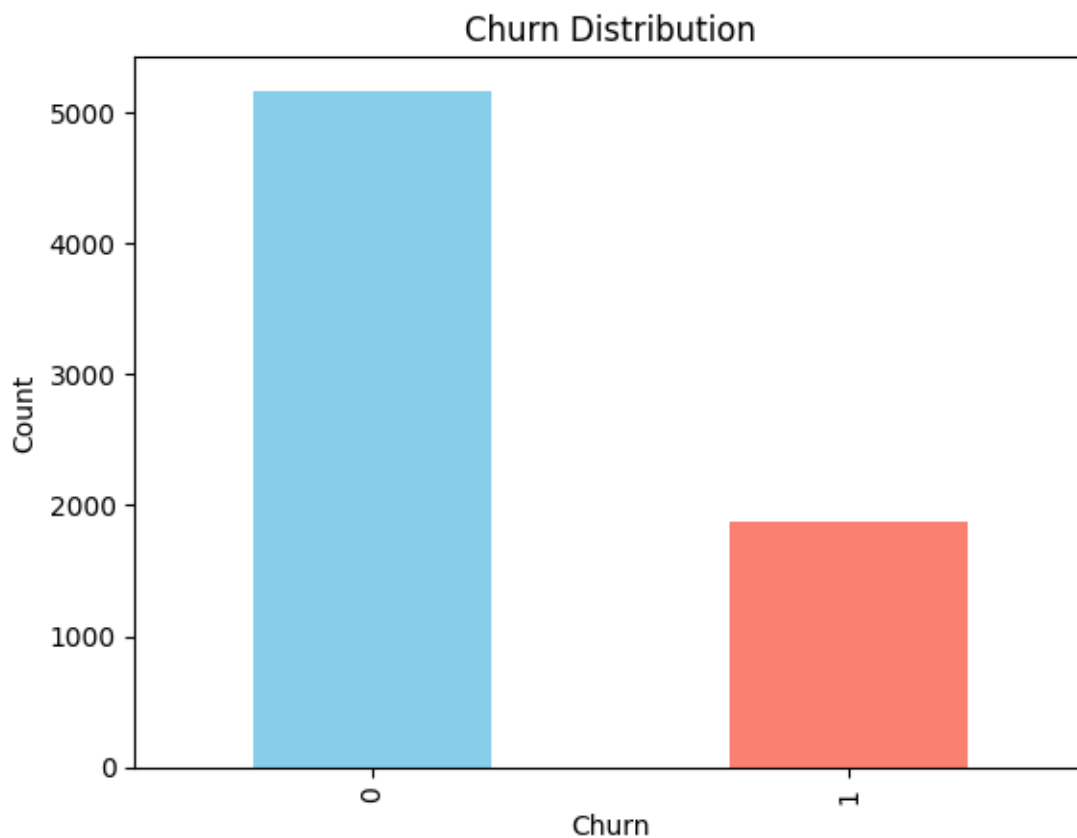
```
print("Dataset Shape:", df.shape)
```

Dataset Shape: (7043, 21)

6. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was conducted to gain insights into the dataset and understand customer churn behavior. The churn distribution analysis showed that the dataset is imbalanced, with a higher number of non-churn customers compared to churn customers. This imbalance highlights the importance of selecting suitable evaluation metrics beyond accuracy.

EDA helped in identifying important patterns and relationships between customer features and churn behavior.



7. METHODOLOGY

7.1 Data Cleaning

The TotalCharges column contained missing and non-numeric values. These values were converted to numeric form, and invalid rows were removed to ensure data quality.

```
# Convert TotalCharges to numeric
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

# Drop only rows where TotalCharges is NaN
print("Rows before cleaning:", df.shape[0])
df = df[df['TotalCharges'].notna()]
print("Rows after cleaning:", df.shape[0])

# Map Churn to numeric
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})
```

[4] Python

Rows before cleaning: 7043
Rows after cleaning: 7032

7.2 Feature Encoding

Categorical features such as gender, contract type, and payment method were converted into numerical form using one-hot encoding to make them suitable for machine learning models.

```
df = pd.get_dummies(df, drop_first=True)
print("After Encoding Shape:", df.shape)
```

[7] Python

After Encoding Shape: (7032, 31)

```
df.head()
```

[8] Python

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges	Churn	gender_Male	Partner_Yes	Dependents_Yes	PhoneService_Yes	Multi-Line_Yes
0	0	1	29.85	29.85	0	False	True	False	False	False
1	0	34	56.95	1889.50	0	True	False	False	True	False
2	0	2	53.85	108.15	1	True	False	False	True	False
3	0	45	42.30	1840.75	0	True	False	False	False	False
4	0	2	70.70	151.65	1	False	False	False	True	False

5 rows × 31 columns

7.3 Feature Scaling

Feature scaling was performed using StandardScaler to normalize numerical features and improve model performance.

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Python

7.4 Train-Test Split

The dataset was split into 80% training data and 20% testing data using stratified sampling to maintain class distribution.

```
X = df.drop('Churn', axis=1)
y = df['Churn']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
print("Train shape:", X_train.shape, "Test shape:", X_test.shape)
```

Python

... Train shape: (5625, 30) Test shape: (1407, 30)

7.5 Machine Learning Models

The following classification models were implemented:

- Logistic Regression (baseline model)
- Random Forest Classifier
- Gradient Boosting Classifier

8. MODEL TRAINING AND EVALUATION

Each model was trained using the training dataset and evaluated on the test dataset. The evaluation metrics included accuracy, precision, recall, F1-score, and confusion matrix.

- Logistic Regression achieved approximately 72.6% accuracy

```
[11]
...
===== Logistic Regression =====
Accuracy: 72.64 %
      precision    recall  f1-score   support

     0       0.91      0.70      0.79      1033
     1       0.49      0.80      0.61       374

   accuracy          0.73      1407
  macro avg       0.70      0.75      0.70      1407
 weighted avg       0.79      0.73      0.74      1407
```

- Random Forest achieved approximately 77.1% accuracy

```
...
===== Random Forest =====
Accuracy: 77.11 %
      precision    recall  f1-score   support

     0       0.89      0.79      0.84      1033
     1       0.55      0.72      0.63       374

   accuracy          0.77      1407
  macro avg       0.72      0.76      0.73      1407
 weighted avg       0.80      0.77      0.78      1407
```

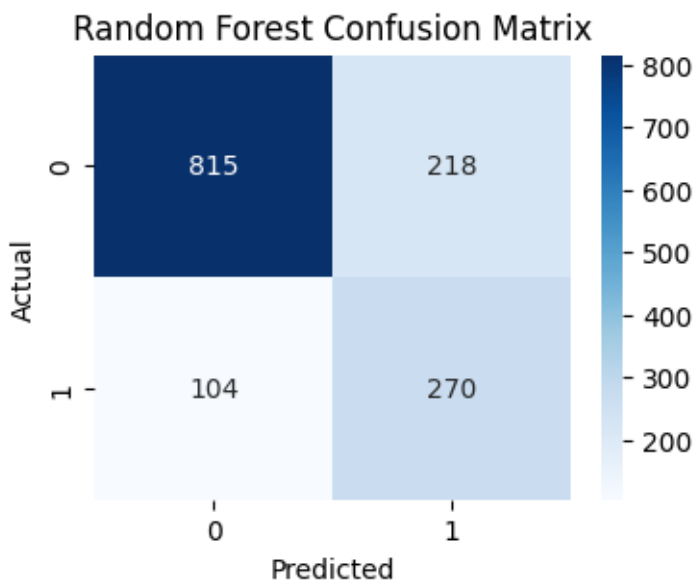
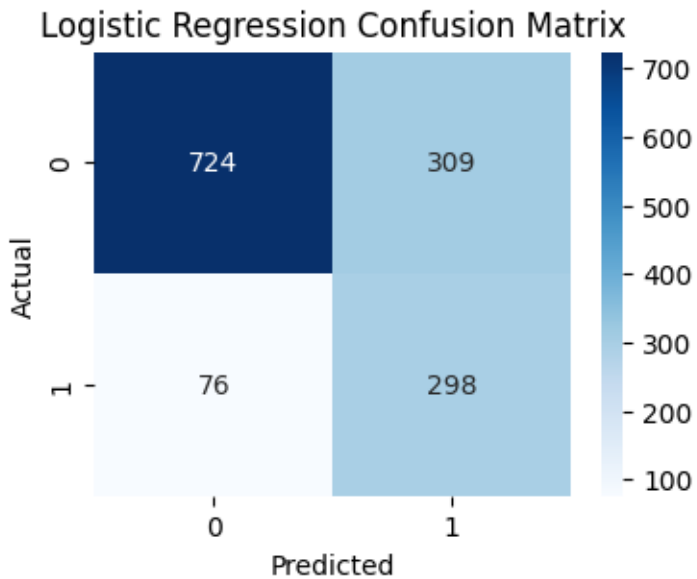
- Gradient Boosting achieved approximately 79.6% accuracy

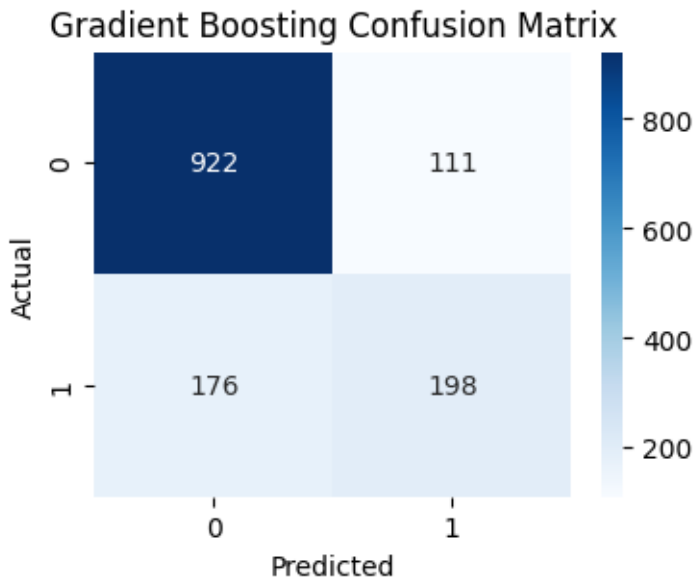
```
...
===== Gradient Boosting =====
Accuracy: 79.6 %
      precision    recall  f1-score   support

     0       0.84      0.89      0.87      1033
     1       0.64      0.53      0.58       374

   accuracy          0.80      1407
  macro avg       0.74      0.71      0.72      1407
 weighted avg       0.79      0.80      0.79      1407
```

Confusion matrices were used to analyze model performance in terms of correct and incorrect predictions.



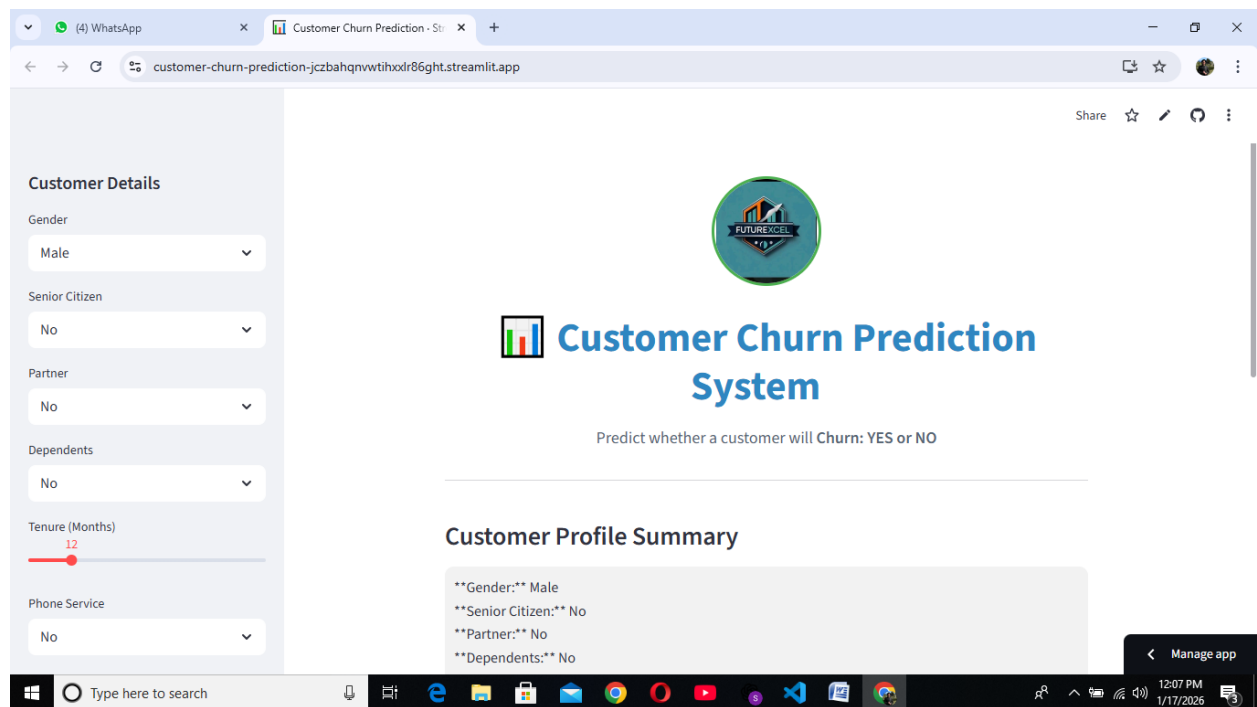


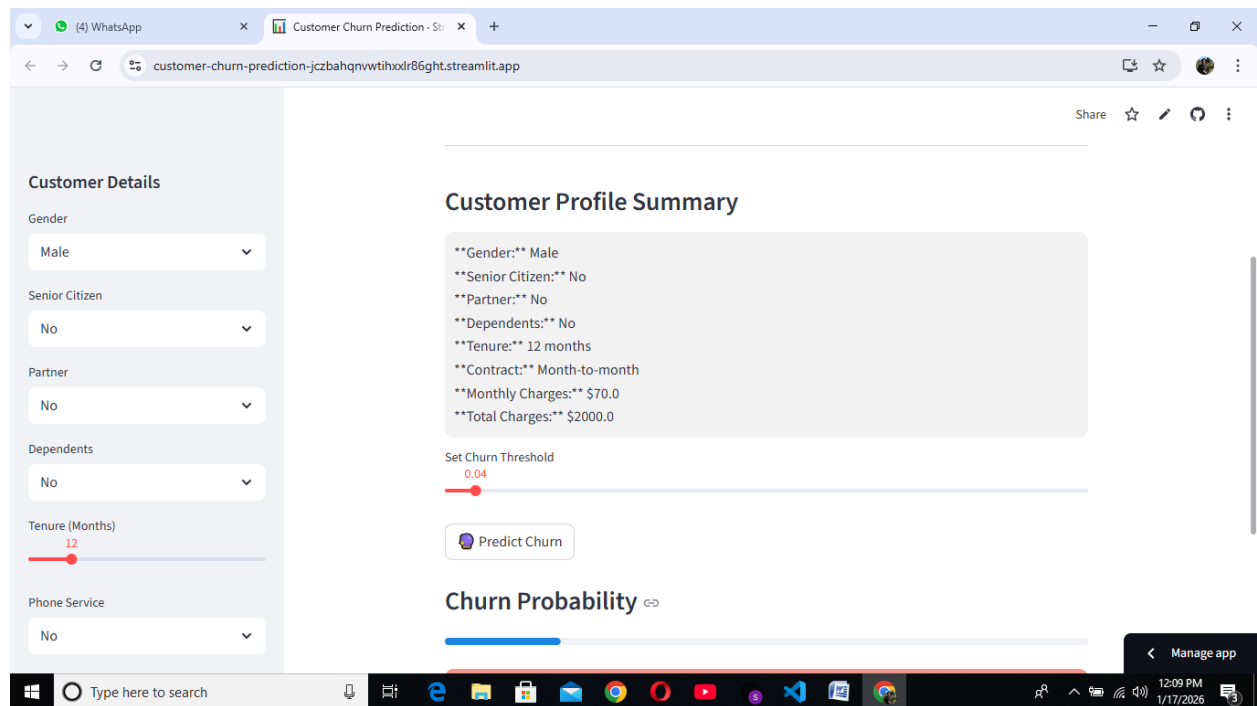
9. FINAL MODEL SELECTION

Gradient Boosting was selected as the final model because it achieved the highest accuracy and demonstrated balanced performance between churn and non-churn classes. Its ability to combine multiple weak learners resulted in improved generalization compared to other models.

10. SYSTEM IMPLEMENTATION (STREAMLIT WEB APPLICATION)

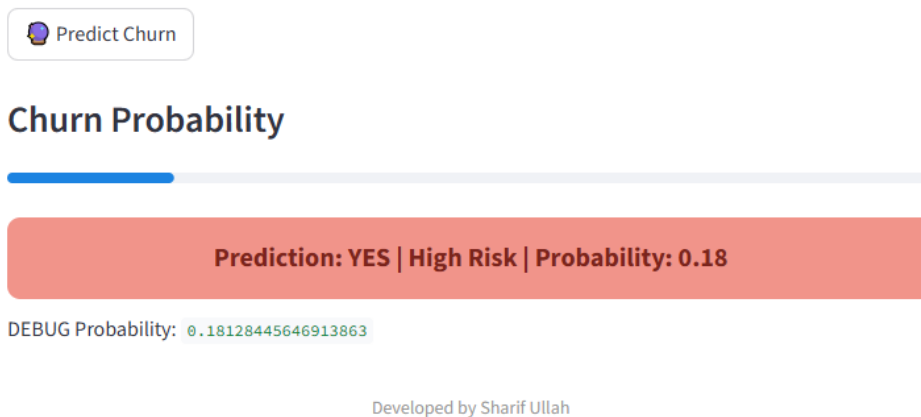
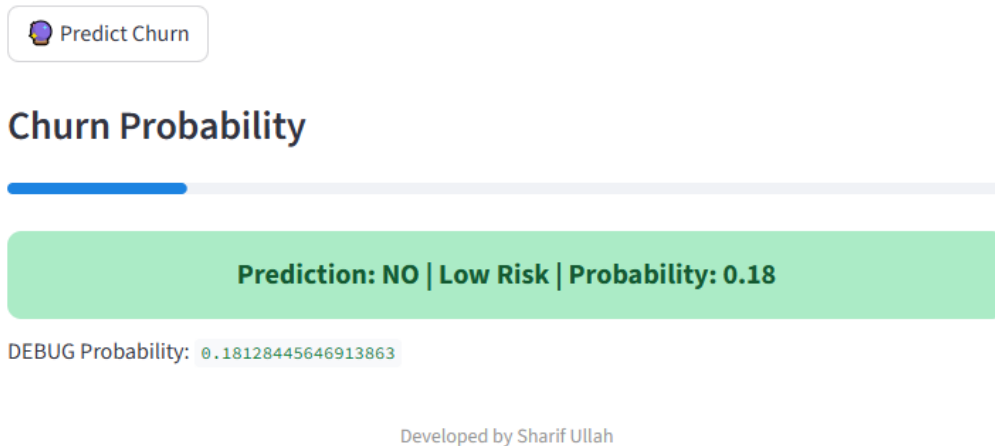
A Streamlit-based web application was developed to deploy the final churn prediction model. The application allows users to input customer information through an interactive interface and receive churn predictions in real time along with probability and risk level. This system enhances usability and practical deployment.





11. RESULTS AND DISCUSSION

The results indicate that customer tenure, contract type, and monthly charges are significant factors influencing churn. The developed system successfully predicts churn with reasonable accuracy and can be used to support customer retention strategies.



12. CONCLUSION

This project successfully demonstrates the application of machine learning techniques for customer churn prediction. By analyzing historical customer data and applying advanced classification models, a reliable churn prediction system was developed. The integration of a web application further enhances the system's practical usability in real-world business scenarios.

13. FUTURE WORK

- Implement advanced algorithms such as XGBoost or deep learning models
- Perform hyperparameter tuning to further improve accuracy
- Integrate real-time customer data
- Deploy the application on cloud platforms

13. REFERENCES

1. Scikit-learn Documentation
2. Kaggle – Telco Customer Churn Dataset
3. Tom Mitchell, *Machine Learning*